

# Yazan Alatrach

yazanalatrach.com

## Experience

### Analytics Engineer - Octopus Energy

Sep '24 - Present

- Owned transformation layer architecture using dbt and SQL to build scalable data models supporting financial reporting, energy trading operations, and risk management across international regions, processing millions of daily events from transactional databases, API logs, and operational systems.
- Built end-to-end automated QA pipeline using Python, Gemini LLM, and Google Drive API to ingest phone transcripts, generate AI-powered quality assessments with constructive feedback, and produce automated reports for energy specialists, eliminating manual review processes and accelerating team training cycles.
- Established technical standards and best practices for data modeling, development, testing, and monitoring, implementing robust change management, source control, and code review processes to ensure data quality and cent-level precision for finance teams.
- Refactored obscure legacy code into a transparent, maintainable system using Python and dbt, reducing processing time by 80% and improving overall system reliability for business-critical energy market operations.
- Enabled cross-functional stakeholders by collaborating directly with finance, operations, and risk management teams to understand requirements and deliver reliable data products via BI tools and operational dashboards, supporting real-time decision-making.
- Designed and implemented 10+ automated data ingestion pipelines using Python and Apache Airflow to integrate critical third-party data sources (Yes Energy, Amperon, Meteologica, Habitat Energy) into AWS data lake, enabling risk teams to perform accurate energy procurement, pricing forecasts, and load forecasts for multi-million dollar trading decisions.
- Built comprehensive monitoring and alerting systems for data pipelines to quickly identify failures and data anomalies, ensuring stability and data quality across the transformation layer.
- Maintained 99%+ uptime of business-critical data systems by proactively monitoring Slack alerts, investigating pipeline failures (expired certificates, rate limiting, incorrect data types), and implementing fixes to prevent disruption to finance, risk, and operations teams relying on real-time data.
- Engineered dynamic customer-to-battery portfolio assignment system using Python to conduct A/B testing, validating that battery-enabled portfolios generated measurably higher profitability compared to non-battery populations, directly informing strategic energy storage investment decisions.

### Senior Data Engineer - Capital One

Aug '21 - May '23

- Built and maintained transformation layer data models using SQL and Snowflake to support diverse business needs, from daily operational metrics to historical trend analysis, enabling stakeholders across finance and operations to access critical insights.
- Optimized critical business queries, reducing query execution time by 50% from 1 hour to 30 minutes, enabling swifter access to key metrics like application submissions, conversion, and fund rates, thereby accelerating daily and historical data analysis.
- Developed robust ETL/ELT pipelines using Snowflake and Apache Airflow to integrate data from CRMs, payment systems, and transactional databases, establishing repeatable patterns for data ingestion and transformation.
- Collaborated directly with data analysts and business users to understand requirements, create and refine Tableau dashboards, and deliver actionable data products that elevated reporting quality and supported informed decision-making.

## Education

Georgia Institute of Technology  
The University of Texas at Austin

M.S Computer Science '25  
B.S Electrical/Computer Engineering '19

## Skills and Certifications

AWS Certified Solutions Architect Associate

Languages: SQL, Python, Unix/Linux, Javascript, Swift

Tools/Frameworks: DBT, AWS, Airflow, Databricks, Streamlit, Snowflake, Jenkins, GCP, Tableau, FastAPI